

Fault Detection in Transmission line using ANN

Supriya Kumari
EED, MANIT
Bhopal, India-462003
skdav.crp@gmail.com

Anchal Mishra
EED, MANIT
Bhopal, India-462003
anchalmishra4081@gmail.com

Ayush Singhal
EED, MANIT
Bhopal, India-462003
ayushsinghal341@gmail.com

Vikram Dahiya
EED, MANIT
Bhopal, India-462003
vikramdahiya646@gmail.com

Mayank Gupta
EED, MANIT
Bhopal, India-462003
mkgupta87988@gmail.com

Suresh Kumar Gawre
EED, MANIT
Bhopal, India-462003
sgawre28@gmail.com

Abstract— Transmission line plays a very important role in ensuring the continuous supply of electricity. However, there is no such thing as perfect or fault-free. Due to various reasons, may that be exposure to the environment, accident, transmission failure or faults due to mis operation; faults occur in transmission line. This results in sudden increase of current, more heat production in conductors leading to some major accidents. The objective of this paper is to fault detection by introducing Artificial Neural Network (ANN) which is not only fast but also very efficient. Initially, the neural network is feed with six input values, i.e., the values of current and voltages of three phases respectively, and trained using these six inputs. And when the fault occurs, the value of pre-fault and post-fault condition are different, the trained ANN compares these pre-fault and post-fault values to give us the result of its deduction. We have used MATLAB to implement ANN for fault detection.

Keywords— Artificial Neural Network; Transmission line; Line-Line fault; Line-Ground fault; Line-Line-Ground fault; Fault Detection, MATLAB.

I. INTRODUCTION

Power systems are primarily made up of three parts: generation, transmission, and distribution. The transmission line however, is most susceptible to fault due to its greater exposure to the environment because it connects the generation unit with the distribution unit.

Therefore, it is crucial to have a well-coordinated and secure system to identify faults. There are usually faults in High Voltage Transmission lines, so it is essential to have a system that can detect and locate the fault with minimal human interference because these lines are long and run through a variety of geographies, so it would take time and be dangerous to physically inspect them. [1-3].

During fault, there is abnormal increase of current in the transmission line and hence the heat generation in conductor increase abnormally, which not only lead to major loss of power but can also cause major accident and damage to the power system equipment. That is why, researcher came up with different types of method for fault detection. This analysis highlighted three important things that are very important, that are, detection, classification and location. Some method for fault detection and classification are artificial neural network (ANN), fuzzy logic, support vector machine (SVM) and others. Each one of these have its own pros and cons [4].

In addition to increasing system reliability, intelligence-based relays would save time and money. As a result, these devices are widely used to detect and locate

faults. Most power systems use artificial neural networks and fuzzy logic systems [3].

Many studies have investigated the use of ANNs in power systems for protection. The use of artificial neural networks for fault detection was studied by E.O Schweitzer and others. Sobajic and Pao used ANN for critical clearing time prediction. Location faults in EHV using ANN has been proposed by Bouthiba [4].

This study is focused on analyzing fast and efficient method for detection of fault and its classification using Artificial Neural Network (ANN). Based on the ANN training, we can increase the efficiency and accuracy of fault detection. In the study, the phase voltages and phase currents are taken as input to the neural network and different faults are also taken in consideration [5-9]. As a method of training neural networks, back-propagation neural networks are also used. The use of ANN allows the improvement of the failure system as the neural network can keep training itself on-line and improve itself by learning and avoid false alarms [6] [10-11]. There are a variety of transmission line configurations available, such as short lengths, long lengths, single circuits and double circuits [1][12]. Electrical power system conditions vary from distribution to distribution. This dynamic change in power system can therefore be incorporated by neural networks [5].

The paper consists of following sections: Section 2 discuss about the reasons as for why the fault occurs in transmission line, Section 3 describes types of faults that can occur, Section 4 and 5 discuss about ANN and how faults are detected using ANN, Section 6 is for the simulation and lastly Section 7 gives us a brief about the preventive measures that can be taken to avoid faults in transmission line.

II. OCCURANCE OF FAULTS IN TRANSMISSION LINE

In this modern era where life cannot be imagined without power, it becomes very important to know the reasons and diagnose faults in transmission line.

Fault is unwanted or abnormal change in the condition of transmission between phases or ground. This can be SLG fault, LL fault or three phase faults.

There are number of possible reasons for occurrence of faults in the transmission line. Some of them are listed as follows:

- 1) Excessive internal and external stresses on the conductors.
- 2) Equipment failures such as transformers, motors, insulation failure, aging, etc.

- 3) Human errors such as improper rating of equipments, witching the circuit while servicing and so on.
- 4) Smoke of fires as due to smoke particles air ionizes.
- 5) Weather conditions like heavy rains, lightning, tower collapsing due to winds, etc.
- 6) Pollution flashover.

III. CLASSIFICATION OF FAULTS

Faults is classified into two types: closed circuit faults and open circuit faults.

A. Closed Circuit Faults

This type of fault is caused due to low impedance path between two points with different voltage values. This results in abnormal increase of current in the transmission line.

The major reason for this type of fault is insulation failure, damage due to some external reason, lightening, etc.

1) Symmetrical Faults

The faults which are symmetrical in nature like L-L-L-G are called symmetrical faults. These are the most severe fault that occur in a transmission line and its prevalence is about 2-3%. [13-15]

2) Unsymmetrical Faults

Faults which usually include only one or two phase are called unsymmetrical faults. The details of these unsymmetrical faults are as follow:

a) *Single Phase to Ground (L-G) Fault:* In this type of fault one phase is short circuited to ground, as shown in Figure 1. The chance of its occurrence or its prevalence is 70-80%. It is the least severe closed circuit fault [13-15].

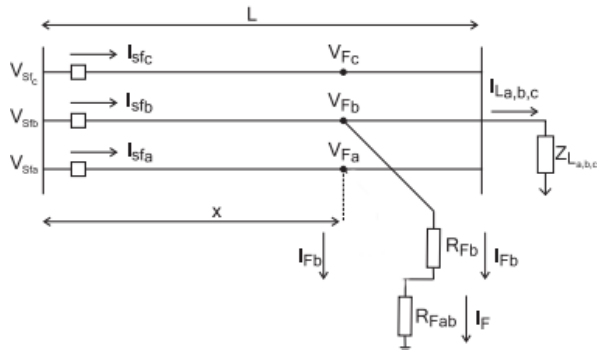


Figure 1: Line-to-ground fault [8]

b) *Two Phase to Ground (L-L-G) Fault:* In this type of fault two phases are short circuited to ground, as shown in Figure 2. The chance of its occurrence or its prevalence is 10-17%. It is more severe than the L-G fault [13-15].

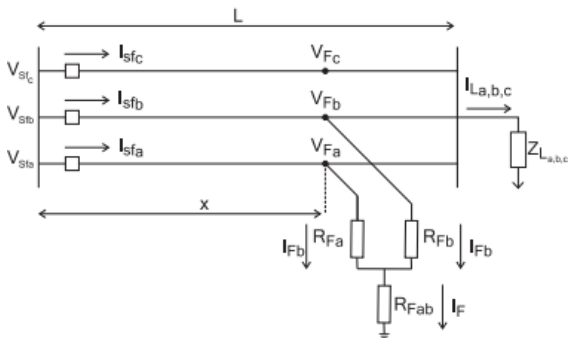


Figure 2: Line to line to ground fault [8]

c) *Phase to Phase (L-L) Fault:* In this type of fault any two phases of the system is short circuited, as shown in Figure 3. The chance of its occurrence or its prevalence is 8-10%. It is more severe than L-L-G fault [13-15].

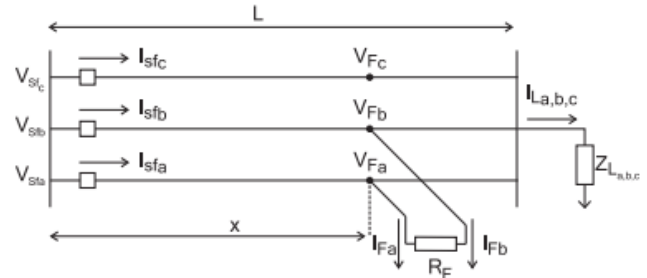


Figure 3: Line to line fault [8]

d) *Phase to Phase to Phase (L-L-L) Fault:* In this type of fault all three phases of the system is short circuited, as shown in Figure 4. Its prevalence is 2-3%. It is the most severe unsymmetrical fault [13-15].

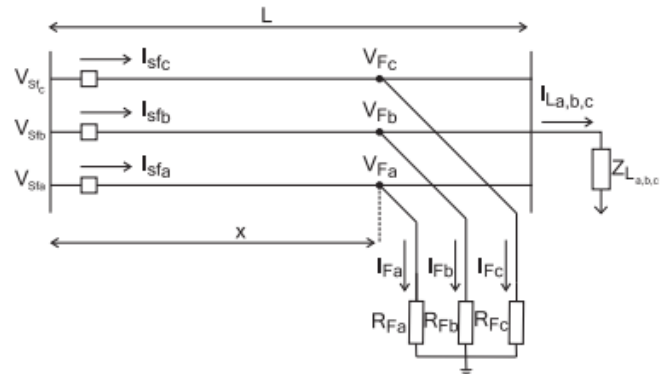


Figure 4: Line-to-line-to-line fault [8]

B. Open Circuit Faults

This type of fault occurs due to conductor breakdown. Such situation happens due to failure to close in phases but the isolators are open. This causes the unbalanced current to flow through system and hence heating the machines [13-15].

IV. ARTIFICIAL NEURAL NETWORK

Artificial Neural Network, commonly known as ANN, is a subset of machine learning. Its name and origin are inspired by human brain. It mimics the way biological neuron work to store and process the data to give the relevant output.

There are node layers, each containing a layer for input, a hidden layer, and a layer for output. A node, or artificial neuron, is connected to another node and has a threshold value associated with it, as shown in Figure 5.

A node is turned on if its output exceeds the specified value, sending data to the next layer. There is no data transfer between layers if this is not the case.

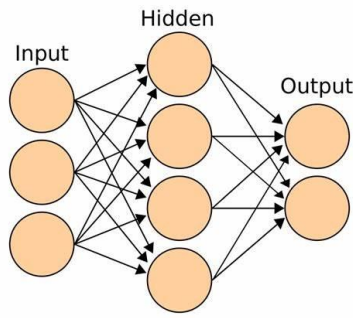


Figure 5: Standard Neural Network [2]

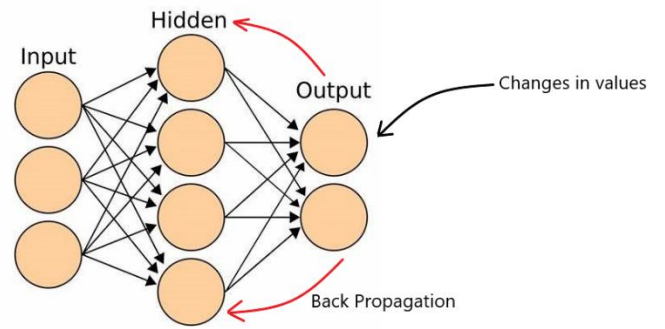


Figure 6: Back Propagation in Neural Network

V. FAULT DETECTION USING ANN

Depending on the type of fault that has occurred in the transmission line, the value of pre fault and post fault voltage and current are very different.

As a result, this method of fault classification combines patterns of pre-fault and post-fault voltages and currents in order to determine the type of fault.

The neural network measures the voltages and currents in the respective three phases of total six inputs and trained using these six inputs. [16-20].

TYPE OF FAULT	A	B	C	G
A-G	1	0	0	1
B-G	0	1	0	1
C-G	0	0	1	1
A-B	1	1	0	0
A-C	1	0	1	0
B-C	0	1	1	0
A-B-G	1	1	0	1
A-C-G	1	0	1	1
B-C-G	0	1	1	1
A-B-C-G	1	1	1	1
NO FAULT	0	0	0	0

Table 1: The binary data values of fault [4]

In total, the neural network gives four output- three phases A, B, C and a ground. The binary data value of these different values is shown in Table 1, by analyzing these values we can find the type of fault occurred in the transmission line.

Back Propagation Neural Network

When it comes to the way neurons are connected, there are two distinct network topologies: feed-forward and feed-back network.

Back Propagation Neural Network (BPNN) are feed-forward networks that forward inputs through intermediate layers until outputs are generated. The outputs are then compared to the desired output in order to determine the errors that will be transmitted backward to the intermediate layer that directly contributes to the output, shown in Figure 6. In order to integrate dynamic changes in power systems into the protection system, Back Propagation Neural Networks are used.

Neural Network learns from observing data sets. After every disturbance, the power system conditions change. The neural network incorporates these changes in the power system. The system obtains knowledge by adapting to the internal and external parameters of the environment and solving complex problems that are difficult to manage [9]. By generalizing knowledge, it is able to respond appropriately to unknown situations. To arrive at solution, data samples are used rather than entire data sets, which is time and cost-effective.

VI. FAULT DETECTION USING SIMULINK MODEL

We developed a SIMULINK model which is single line diagram of a Power System with 11kV/30MVA generating station and a step-down transformer of rating 1MVA, 11kV/0.4kV. Now with the help of “Three-phase Fault” block we obtain the required faults. Various types of faults were simulated along the 100-km line at varying locations and with different fault resistance using a three-phase fault simulator.

Faults on transmission line can be classified as single line to ground fault (LG fault), line to line fault (LL fault), double line to ground fault (LLG fault) and three phase faults (LLL or LLLG fault). In order to model the line, distributed parameters were used, and these parameters were derived from the transmission line was modelled and simulated using the SimPower System library in MATLAB.

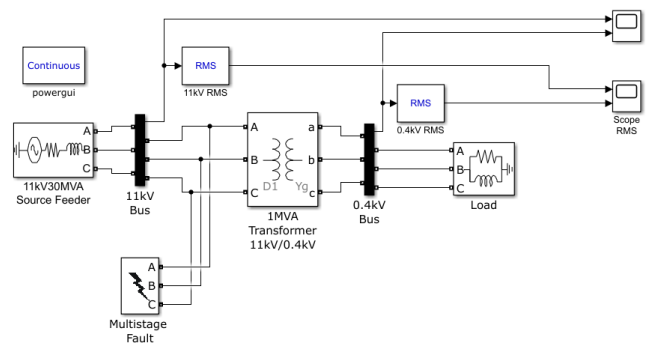


Figure 7: Transmission line Simulation

The data from this simulation is obtained and used to train and feed the test data set to the neural network.

At the terminal, voltage and current samples were measured using the three phase V-I measurement block as shown in Figure 7.

Fault Study using the MATLAB Simulation

The flowchart of testing algorithm is shown in Figure 8.

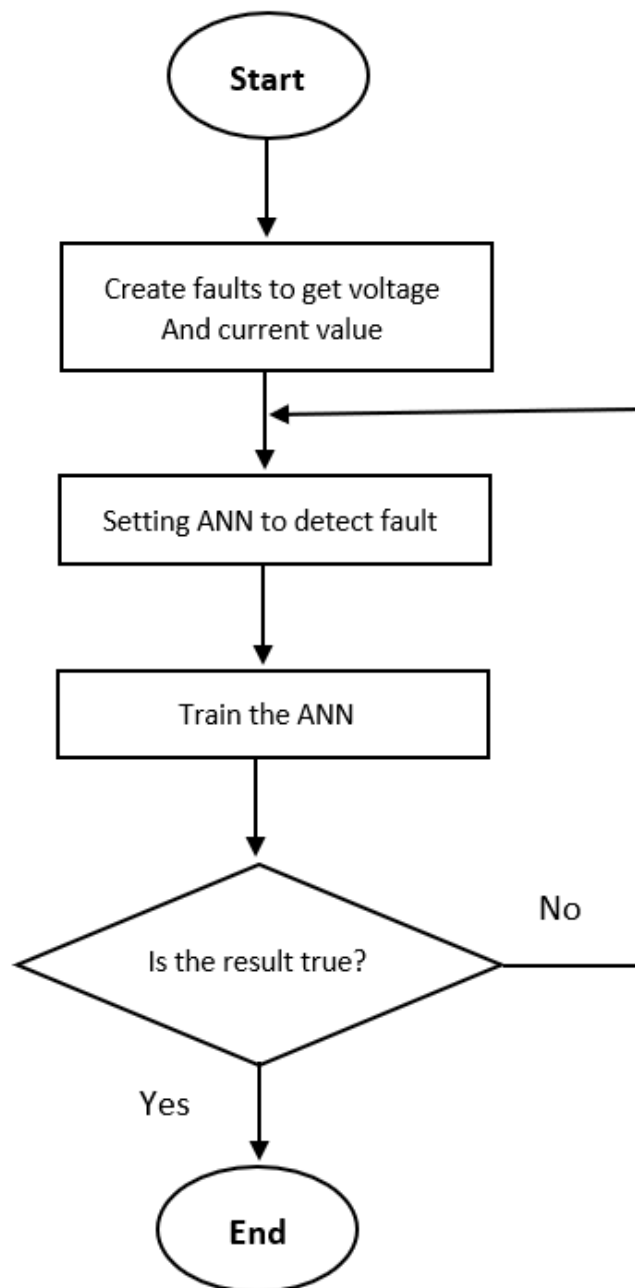


Figure 8: Flowchart of the simulation

For instance, if fault is generated between phases B, C and ground (Figure 9 shows the fault voltage and current graph). After simulation the data set obtained for the fault shown as graph in Figure 10.

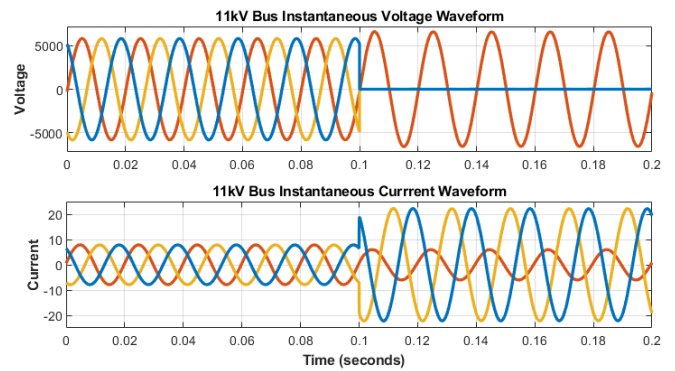


Figure 9: Graph of Voltage and Current during Fault

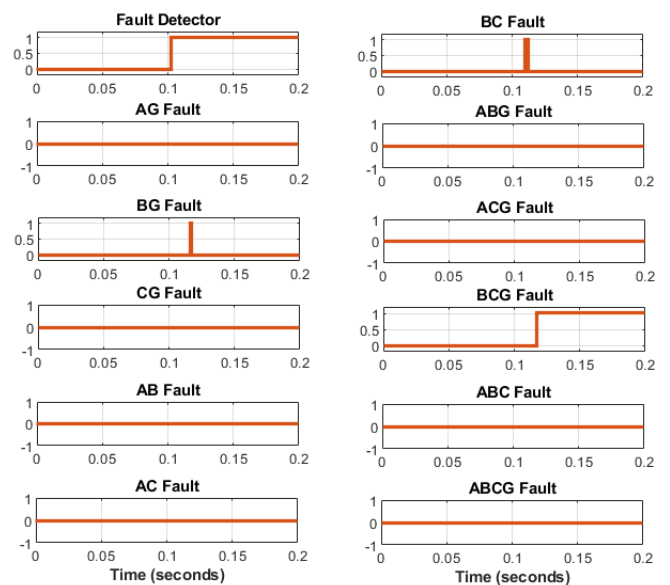


Figure 10: Fault Detection Graph

As shown in Figure 10, the fault detector changes from 0 to 1 for fault indication i.e., there is a fault in the given transmission line. For fault classification the value changed from 0 to 1 indicates the fault in the given phase and if the result remains 0, it means there is no fault.

VII. PREVENTIVE MEASURES FOR FAULTS IN TRANSMISSION LINES

After detection and fixing of Fault in transmission lines, it becomes an important task to have some preventive measures against the faults to ensure that the fault do not readily occur in nearby future. These protective measures depend on the reasons due to which fault occurred:

1) *Prevention against Lightning:* This can be achieved by the use of anti-winding lightning rods, high performance insulation materials to enhance the insulation capacity etc.

2) *Prevention against Tower Collapsing:* While installing the tower, some factors should be kept in mind like ground that are not solid are easy to collapse, maintenance inconvenient should be avoided.

3) *Prevention from Pollution:* Insulation materials on overhead transmission lines should be inspected at time to time to maintain their good working and operation status should be monitored timely using online monitoring.

4) *Prevention from Equipment Failures*: This is the major reason of faults generation. It can be limited using fault limiting devices like fuse, circuit breakers, relays, etc.

VIII. CONCLUSION

This paper depicts better fault detection and its classification techniques for the power system. It is reflected that the use of Artificial Neural Network (ANN) for the detection and classification of fault in power system has been thoroughly investigated. Along with that causes, types of faults and working of ANN is thoroughly discussed. When ANN based fault detection system is used along with protective circuits, many major accidents can be avoided. Faults lead to the loss of equipment as it may burn the equipment. So, costly equipment needs to be protected. The use protection circuit alongside ANN, minimizes these losses. As the ANN would quickly and efficiently detect the fault and can switch on the protection circuit with least human interference. Some preventive measures should also be taken to avoid further inconvenience and smooth transmission of electricity.

REFERENCES

- [1] Das R, Novosel D, "Review of fault location techniques for transmission and sub-transmission lines," Proceedings of 54th Annual Georgia Tech Protective Relaying Conference, 2000.
- [2] N. A. M. Leh, F. M. Zain, Z. Muhammad, S. A. Hamid and A. D. Rosli, "Fault Detection Method Using ANN for Power Transmission Line," 10th IEEE International Conference on Control System, Computing and Engineering (ICCSCE), pp. 79-84, doi: 10.1109/ICCSCE50387.2020.9204921, 2020.
- [3] Magnago FH, Abur A, "Advanced techniques for transmission and distribution system fault location," Proceedings of CIGRE – Study committee 34 Colloquium and Meeting, Florence, paper 215, 1999.
- [4] M. A. Alashter, O. G. Mrehel and A. S. Shamekh, "Design and Evaluation a Distance Relay Model Based On Artificial Neural Networks (ANN)," 6th IEEE International Energy Conference (ENERGYCon), pp. 685-690, doi: 10.1109/ENERGYCon48941.2020.9236490, 1999.
- [5] Jamil, M., Sharma, S.K. & Singh, R. "Fault detection and classification in electrical power transmission system using artificial neural network," *SpringerPlus* 4, 334 (2015).
- [6] S. Naidu, E. Zafiriou and T. J. M. Avoy, "Application of Neural Networks on the Detection of Sensor Failure During the Operation of a Control System," American Control Conference, pp. 1336-1341, doi: 10.23919/ACC.1989.4790398, 1989.
- [7] R. Resmi, V. Vanitha, E. Aravind, B. R. Sundaram, C. R. Aswin and S. Hariithaa, "Detection, Classification and Zone Location of Fault in Transmission Line using Artificial Neural Network," IEEE International Conference on Electrical, Computer and Communication Technologies (ICECCT), pp. 1-5, doi: 10.1109/ICECCT.2019.8868990, 2019.
- [8] Filomena, André D.. & Bretas, Arturo S., "Distribution systems fault analysis considering fault resistance estimation," *International Journal of Electrical Power & Energy Systems*, 33(7), 1326-1335, 2011. <https://doi.org/10.1016/j.ijepes.2011.06.010>
- [9] Ravi V. Ghodchar, Dr. R. G. Karandikar, "Fault Detection and Classification in Power Transmission Lines Based on Transient Signal Analysis: A Review of Different Methodologies," *International Journal of Engineering Research & Technology (IJERT)* Volume 02, Issue 11 (November 2013),.
- [10] Patel, Mamta & Patel, Ramnarayan, "Fault Detection and Classification on a Transmission Line using Wavelet Multi Resolution Analysis and Neural Network," *International Journal of Computer Applications*. 47. 10.5120/7489-0543, 2012.
- [11] F. Fayaz, A. Isa, H. K. Verma and S. Deb, "Improved ANN-based algorithm for detection and classification of faults on transmission lines," 1st India International Conference on Information Processing (IICIP), 2016, pp. 1-6, doi: 10.1109/IICIP.2016.7975360, 2016.
- [12] Seada Hussien Adem, "Fault Detection, Protection and Location on Transmission Line," A Review, Munich, GRIN Verlag, 2020.
- [13] M. R. Bishal, S. Ahmed, N. M. Molla, K. M. Mamun, A. Rahman and M. A. A. Hysam, "ANN Based Fault Detection & Classification in Power System Transmission line," *International Conference on Science & Contemporary Technologies (ICSCT)*, 2021, pp. 1-4, doi: 10.1109/ICSCT53883.2021.9642410, 2021.
- [14] K. Hasija, S. Vadhera, A. Kumar and A. Kishore, "Detection and location of faults in underground cable using Matlab/Simulink/ANN and OrCad". 6th IEEE Power India International Conference (PIICON), 2014, pp. 1-5, doi: 10.1109/POWERI.2014.7117624, 2014.
- [15] S. M. Chopdar and A. K. Koshti, "Fault Detection and Classification in Power System Using Artificial Neural Network," 2nd International Conference on Intelligent Technologies (CONIT), pp. 1-6, doi: 10.1109/CONIT55038.2022.9848016, 2022.
- [16] M. Sanaye-Pasand and O. P. Malik, "Power transmission lines fault direction estimation using artificial neural networks," in *Proceedings of the Canadian Conference on Electrical and Computer Engineering*, vol. 2, pp. 758–761, May 1996
- [17] A. Jain, V. S. Kale, and A. S. Thoke, "Application of artificial neural networks to transmission line faulty phase selection and fault distance location," in *Proceedings of the IASTED International Conference "Energy and Power System"*, pp. 262–267, Chiang Mai, Thailand, 2006
- [18] N. Saravanan and A. Rathinam, "A comparative study on ANN based fault location and classification technique for double circuit transmission line," in *Proceedings of the 4th International Conference on Computational Intelligence and Communication Networks (CICN '12)*, pp. 824–830, November 2012.
- [19] A. Jain, A. S. Thoke, and R. N. Patel, "Fault classification of double circuit transmission line using artificial neural network," *International Journal of Electrical Systems Science and Engineering*, vol. 1, no. 4, pp. 750–755, 2008.
- [20] J. Holkar and P. V. Fulmali, "Fault Analysis on Transmission Lines using Artificial Neural Network," *Int. J. Eng. Sci. Res. Technol.*, vol. 5, no. 2, pp. 863– 871, 2016.